

CLASS 11-CBSE MATHEMATICS
CHAPTER: CONIC SECTIONS-SET-1

TIME : 1Hr

Mx.Marks:30

SECTION A

Q.I Answer the following questions **[1X3 = 3 MARKS]**

1. Find the radius of the circle $2(x^2 + y^2) + 4x - 3y + 1 = 0$.
2. Find the equation of a circle whose Centre is (2, -1) and radius is 3.
3. If the latus rectum of an ellipse be equal to half of its minor axis, then its eccentricity is

SECTION B

Q.II Answer the following questions **[2X5 = 10 MARKS]**

4. If (x, 3) and (3, 5) are the extremities of a diameter of a circle with centre at (2, y). Then find the value of x and y.
5. The vertices of a hyperbola are at (0, 0) and (10, 0) and one of its foci is at (18, 0). The write the equation of the hyperbola.
6. Write the equation of the conic with focus at (1, -1), directrix along $x - y + 1 = 0$ and with eccentricity $\sqrt{2}$.
7. Find the equations of the tangents to the ellipse $3x^2 + 4y^2 = 12$ which are perpendicular to the line $y + 2x = 4$.
8. The distance of a point on the ellipse $\frac{x^2}{6} + \frac{y^2}{2} = 1$ from the centre is 2. Then find the eccentric angle of the point.

SECTION C

Q.III Answer the following questions **[3X3 = 9 MARKS]**

9. The equation $16x^2 - 3y^2 - 32x + 12y - 44 = 0$ represents a hyperbola, what is its eccentricity.



10. Write the parametric representation of a point on the ellipse whose foci are $(-1, 0)$ and $(7, 0)$ and eccentricity $\frac{1}{2}$.

11. Find the product of the perpendiculars from the foci upon any tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

SECTION D

Q.IV Answer the following questions

[4X2 = 8 MARKS]

12. Write the equation of a common tangent with positive slope to the circle $x^2 + y^2 - 8x = 0$ as

well as to the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$?

13. Find the equations of the tangents to the ellipse $4x^2 + 3y^2 = 5$ which are inclined at an angle of 60° to the axis of x. Also, find the point of contact.

